Practical No. 8

Aim: SQLite Database

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout android:layout_height="match_parent"
    android:layout_width="match_parent"
    xmlns:android="http://schemas.android.com/apk/res/android">

    <EditText
        android:id="@+id/edtEnterName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="32dp"
        android:hint="Enter Name"
        android:layout_centerHorizontal="true" />

    <EditText
        android:id="@+id/edtEnterAge"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="32dp"
        android:hint="Enter Age"
        android:layout_below="@id/edtEnterName"
        android:layout_centerHorizontal="true"/>

    <Button
        android:id="@+id/btnAddName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:text="Add Name and Age"
        android:textSize="20dp"
        android:layout_below="@id/edtEnterAge"
        android:layout_centerHorizontal="true"/>

    <Button
        android:id="@+id/btnPrintName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:text="Print Name and Age"
        android:textSize="20dp"
        android:layout_below="@id/btnAddName"
        android:layout_centerHorizontal="true"/>
```

```
    <TextView
        android:id="@+id/tvName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:textSize="20dp"
        android:layout_below="@id/btnPrintName"
        android:layout_centerHorizontal="true" />

    <TextView
        android:id="@+id/tvAge"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20dp"
        android:layout_marginTop="32dp"
        android:layout_below="@id/tvName"
        android:layout_centerHorizontal="true" />

</RelativeLayout>
```

DBHelper.kt

```
import android.content.ContentValues
import android.content.Context
import android.database.Cursor
import android.database.sqlite.SQLiteDatabase
import android.database.sqlite.SQLiteOpenHelper

class DBHelper(context: Context, factory: SQLiteDatabase.CursorFactory?) :
    SQLiteOpenHelper(context, DATABASE_NAME, factory, DATABASE_VERSION) {

    // Called when the database is created for the first time
    override fun onCreate(db: SQLiteDatabase) {
        val createTableQuery = """
            CREATE TABLE $TABLE_NAME (
                $ID_COL INTEGER PRIMARY KEY AUTOINCREMENT,
                $NAME_COL TEXT,
                $AGE_COL TEXT
            )
        """.trimIndent()

        db.execSQL(createTableQuery)
    }

    // Called when the database needs to be upgraded
```

```
    override fun onUpgrade(db: SQLiteDatabase, oldVersion: Int, newVersion: Int) {
        db.execSQL("DROP TABLE IF EXISTS $TABLE_NAME")
        onCreate(db)
    }

    // Inserts a new record into the database
    fun addName(name: String, age: String) {
        val values = ContentValues().apply {
            put(NAME_COL, name)
            put(AGE_COL, age)
        }

        writableDatabase.use { db ->
            db.insert(TABLE_NAME, null, values)
        }
    }

    // Retrieves all records from the database
    fun getName(): Cursor {
        return readableDatabase.rawQuery("SELECT * FROM $TABLE_NAME", null)
    }

    companion object {
        private const val DATABASE_NAME = "ANDROID"
        private const val DATABASE_VERSION = 1
        const val TABLE_NAME = "android_table"
        const val ID_COL = "id"
        const val NAME_COL = "name"
        const val AGE_COL = "age"
    }
}
```

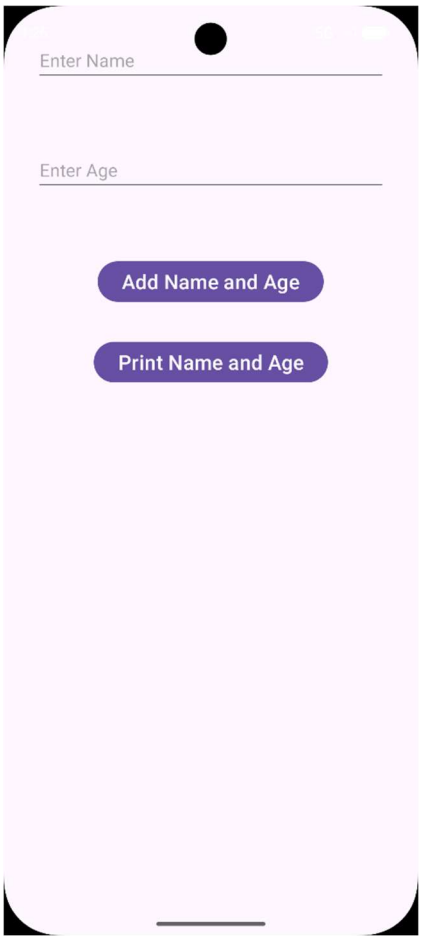
MainActivity.kt

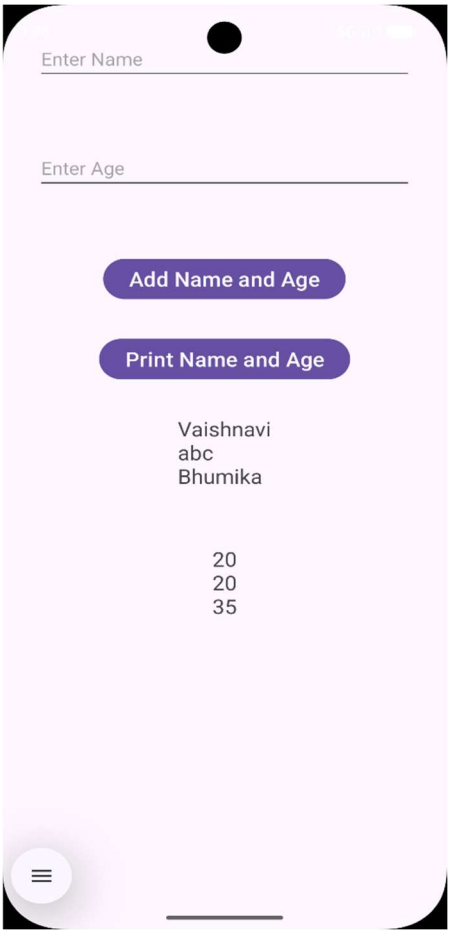
```
package com.example.myapplication

import DBHelper
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
    private lateinit var addName: Button
```

```
    private lateinit var printName: Button
    private lateinit var enterName: EditText
    private lateinit var enterAge: EditText
    private lateinit var name: TextView
    private lateinit var age: TextView

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        // Initialize UI components
        addName = findViewById(R.id.btnAddName)
        printName = findViewById(R.id.btnPrintName)
        enterName = findViewById(R.id.edtEnterName)
        enterAge = findViewById(R.id.edtEnterAge)
        name = findViewById(R.id.tvName)
        age = findViewById(R.id.tvAge)

        // Create database helper instance
        val db = DBHelper(this, null)

        // Add data to database on button click
        addName.setOnClickListener {
            val inputName = enterName.text.toString().trim()
            val inputAge = enterAge.text.toString().trim()

            if (inputName.isEmpty() || inputAge.isEmpty()) {
                // Show message if fields are empty
                Toast.makeText(this, "Please enter both name and age",
Toast.LENGTH_SHORT).show()
            }
            else
            {
                // Add name and age to database
                db.addName(inputName, inputAge)
                Toast.makeText(this, "$inputName added to database",
Toast.LENGTH_LONG).show()

                // Clear input fields
                enterName.text.clear()
                enterAge.text.clear()
            }
        }

        // Retrieve and display data from database on button click
        printName.setOnClickListener {
            name.text = ""
            age.text = ""
```

```
        val cursor = db.getName()

        cursor.use {
            if (cursor.moveToFirst()) {
                do {
                    val personName =
cursor.getString(cursor.getColumnIndexOrThrow(DBHelper.NAME_COL))
                    val personAge =
cursor.getString(cursor.getColumnIndexOrThrow(DBHelper.AGE_COL))

                    // Append data to text views
                    name.append("$personName\n")
                    age.append("$personAge\n")
                } while (cursor.moveToNext())
            }
        }
    }
  }
}
```

Output:


Enter Name
Enter Age
Add Name and Age
Print Name and Age


Enter Name
Enter Age
Add Name and Age
Print Name and Age
Vaishnavi
abc
Bhumika
20
20
35